TANISHQ SAXENA

AI/ML Engineer • Backend Developer • Data Scientist
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PROFILE

Computer Science engineer (JUET, Expected May 2026) specialising in ML systems, Python backend, and data engineering. Deployed a Flask NLP API serving 150+ req/min at 99% uptime; trained a Deep Q-Network to 96% lane-navigation accuracy across 1,000+ episodes; automated reporting pipelines recovering 10+ engineering hours/sprint. LeetCode Top 30% (1531 rating, 200+ problems). SIH 2023 national competitor. Seeking an AI/ML or Backend Engineering role at a high-growth tech startup.

EXPERIENCE

Python Backend Developer — Intern · Octanet (Remote) *May 2025 – Jul 2025*

- Reduced manual deployment overhead by 30% and recovered ~10 engineering hours/sprint for a 5-person team, by engineering 8 automated reporting pipelines in Python and Pandas with modular, reusable components.
- Achieved 100% test coverage and zero production regressions across 12 backend features over 3 Agile sprints, by writing comprehensive unit tests integrated via GitHub Actions CI/CD pipelines.
- Surfaced 4 data-driven business insights that directly shaped 2 sprint backlog decisions, by analysing 15,000+ records using Python (Pandas, SQL) and delivering Matplotlib dashboards within a 48-hour turnaround.
- Improved ML model evaluation score by 25% within 3 weeks — no architecture change — by applying systematic feature engineering and Scikit-learn GridSearchCV hyperparameter tuning on the production pipeline.

PROJECTS

Memory Optimisation via Reinforcement Learning [Live Demo ■]
Python · TensorFlow · Deep Q-Networks · NumPy · System Optimisation

- Reduced memory usage by 28% while maintaining 95% system performance stability across dynamic workloads, by designing a DQN-based agent that learns optimal memory allocation policies end-to-end.
- Improved convergence speed by 40% and saved ~100 compute hrs/year, by engineering a vectorised state-action pipeline handling 500K+ transitions with prioritised experience replay buffers.
- Packaged the trained agent as a reusable Python module with a clean API, enabling drop-in integration into any memory-intensive system with under 10 lines of configuration code.

Self-Driving Car Simulator — Deep RL Agent [Live Demo ■]
PyTorch · Deep Q-Network · OpenAI Gym · Pygame · Python

- Trained DQN agent from scratch to 96% lane-navigation success across 1,000+ episodes; reward redesign and network pruning cut convergence time by 45% with no accuracy loss.
- Built a fully custom Pygame simulation environment — no third-party physics engine — enabling reproducible benchmarking and modular reward shaping for rapid experimentation.
- Implemented epsilon-greedy exploration with decaying schedule and target-network soft updates, stabilising training and eliminating reward oscillation beyond episode 400.

Amazon Sentiment Analysis API [Live Demo ■]
Flask · Python · TF-IDF · Logistic Regression · NLTK · Scikit-learn · REST

- Deployed production Flask REST API classifying 10,000+ reviews at 87% accuracy, live-serving 150+ req/min at 99% uptime — zero cold-start failures across a 30-day production window.
- Eliminated 15 hrs/week of manual preprocessing via automated NLTK tokenisation pipeline, tripling throughput and enabling same-day turnaround for new datasets.
- Engineered structured JSON error responses and rotating request logging, reducing debug time by ~60% and making the endpoint drop-in ready for any product backend.

TECHNICAL SKILLS

Languages	Python, C++, Java, SQL, HTML/CSS
ML / AI	TensorFlow, PyTorch, Scikit-learn, Keras, HuggingFace Transformers, Deep Learning, NLP, Reinforcement Learning, LLM APIs
Data Science	Pandas, NumPy, Matplotlib, Plotly, Statistical Analysis, Predictive Modelling, EDA
Backend	Flask, FastAPI, REST APIs, Streamlit, NLTK, OpenAI Gym
DevOps / Tools	Docker, Git, GitHub Actions, Postman, Jupyter Notebook, VS Code, Linux
CS Fundamentals	Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, DBMS, Operating Systems

EDUCATION

B.Tech — Computer Science & Engineering *2022 – Present*
Jaypee University of Engineering and Technology, Guna, MP ***Expected: May 2026***

ACHIEVEMENTS & CERTIFICATIONS

- **Smart India Hackathon 2023** — Top national competitor among 12,000+ teams; shipped a working analytics product under a 36-hour deadline.
- **LeetCode** — Top 30% globally, Rating 1531; 200+ problems solved across Easy, Medium, and Hard tiers.
- **ML Bootcamp** (40+ hrs) — 6 production NLP & recommendation systems deployed end-to-end.
- **DSA Specialisation — Infosys Springboard**: 65+ challenges, 92% optimisation efficiency; graphs, DP & greedy algorithms.
- **100 Days of Code** — 18 Python apps shipped: web scrapers, automation bots, REST clients, and end-to-end ML demos.